04_phantom_read_demo

Propietario	© Carlos Roberto
Verificación	Verificada
Etiquetas	Arquitectura de Datos

Shows phantom reads under Repeatable Read and how Serializable prevents them.

4.1 Setup

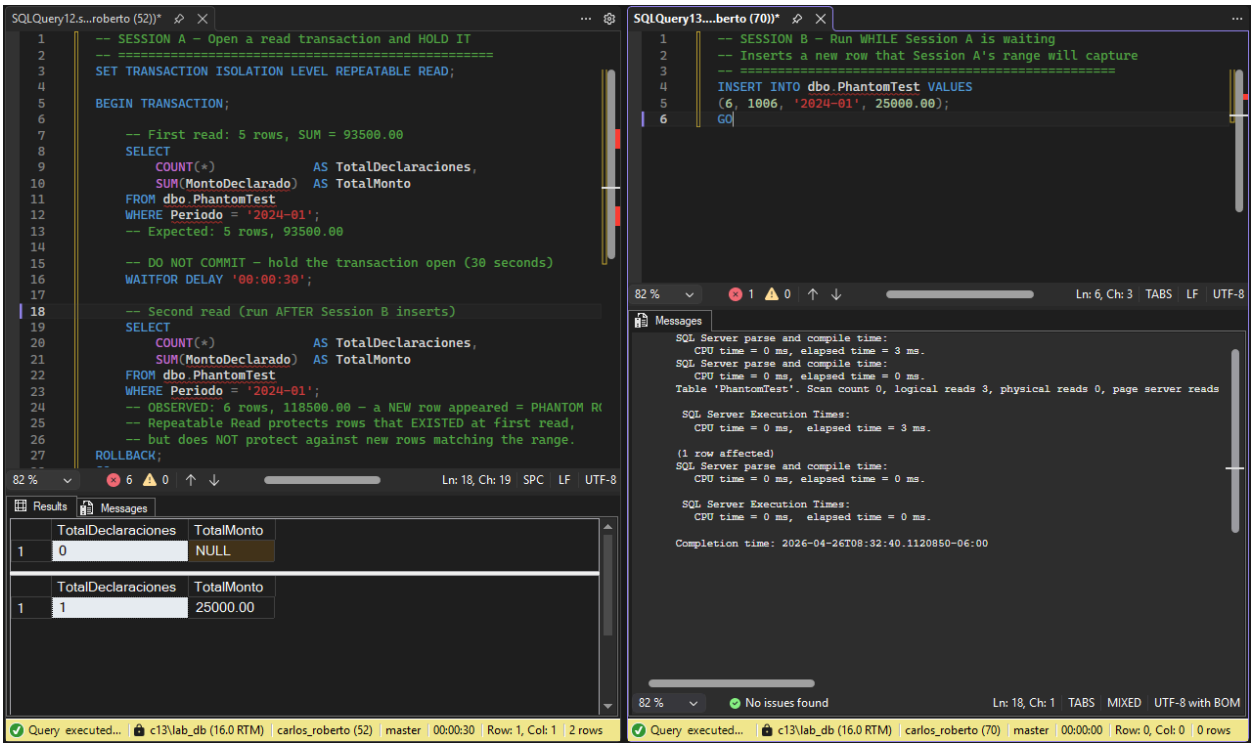
```
USE STN_Lab
GO

-- ==========================================================
==
-- SETUP — Isolated test table for clear observation
-- ==========================================================
==
IF OBJECT_ID('dbo.PhantomTest', 'U') IS NOT NULL DROP TABLE d
bo.PhantomTest
GO

CREATE TABLE dbo.PhantomTest (
    DeclaracionID   INT           NOT NULL PRIMARY KEY,
    ContribuyenteID INT           NOT NULL,
    Periodo         CHAR(7)       NOT NULL,
    MontoDeclarado  DECIMAL(18,2) NOT NULL
)
GO

-- Seed with 5 declarations for period 2024-01
INSERT INTO dbo.PhantomTest VALUES
```

```
(1, 1001, '2024-01', 15000.00),
(2, 1002, '2024-01', 22000.00),
(3, 1003, '2024-01', 8500.00),
(4, 1004, '2024-01', 31000.00),
(5, 1005, '2024-01', 17000.00)
GO
```

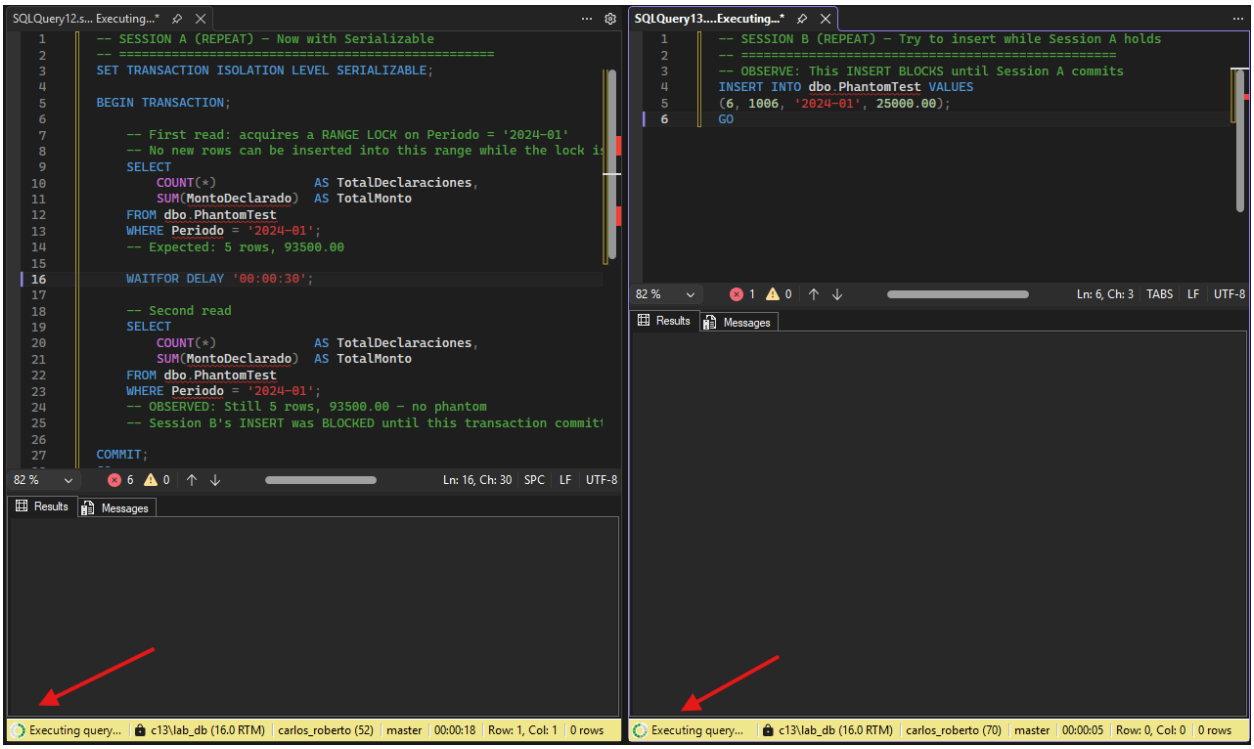
4.2 PHANTOM READ under REPEATABLE READ

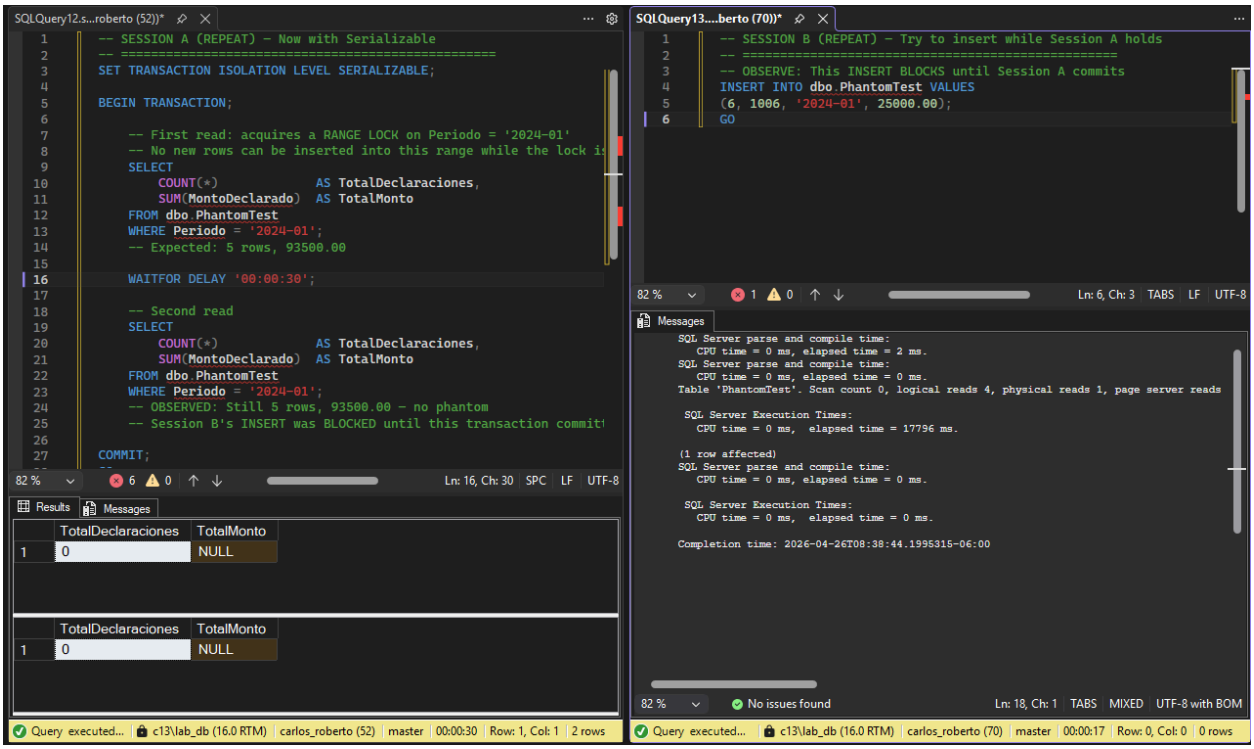
```
-- SESSION A — Open a read transaction and HOLD IT
-- ==================================================
SET TRANSACTION ISOLATION LEVEL REPEATABLE READ;

BEGIN TRANSACTION;

    -- First read: 5 rows, SUM = 93500.00
    SELECT
        COUNT(*)             AS TotalDeclaraciones,
        SUM(MontoDeclarado)  AS TotalMonto
    FROM dbo.PhantomTest
    WHERE Periodo = '2024-01';
    -- Expected: 5 rows, 93500.00

    -- DO NOT COMMIT — hold the transaction open (30 seconds)
    WAITFOR DELAY '00:00:30';

    -- Second read (run AFTER Session B inserts)
    SELECT
        COUNT(*)             AS TotalDeclaraciones,
        SUM(MontoDeclarado)  AS TotalMonto
    FROM dbo.PhantomTest
    WHERE Periodo = '2024-01';
    -- OBSERVED: 6 rows, 118500.00 — a NEW row appeared = PHA
NTOM ROW
    -- Repeatable Read protects rows that EXISTED at first re
```

```
ad,
    -- but does NOT protect against new rows matching the ran
ge.

ROLLBACK;
GO

-- ====================================================
-- SESSION B — Run WHILE Session A is waiting
-- Inserts a new row that Session A's range will capture
-- ====================================================
INSERT INTO dbo.PhantomTest VALUES
(6, 1006, '2024-01', 25000.00);
GO
```



4.3 SERIALIZABLE prevents phantom reads





Observation — Serializable Isolation

Session A holds a range lock on Periodo = '2024-01'. No new rows can be inserted into that range until the transaction finishes.

Session B tried to insert a new row for the same period and got blocked immediately. It waited the entire time Session A was running.

When Session A committed, Session B was unblocked and the insert completed.

The result: Session A read the same count and total both times — no phantom row appeared. Consistency guaranteed.

The cost: Session B was blocked for the full duration of Session A's transaction. In STN, this means no taxpayer can submit a declaration for a period that is currently being totalized.

The architectural rule: Use Serializable only when the consistency requirement is non-negotiable — like fiscal period closing. Never as a default. The throughput cost is too high.

4.4 CLEANUP

```
DROP TABLE IF EXISTS dbo.PhantomTest;
GO
```